Project Title:

🚦 *Traffic Density Analyzer using AI & Machine Learning*

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Course:
CSA2001 – Fundamentals of AI & ML

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Abstract

This project aims to analyze traffic conditions by detecting vehicles in a video and classifying traffic density as low, medium, or high. The system uses computer vision techniques with OpenCV and a pre-trained Haar Cascade classifier to identify vehicles. Based on the number of detected vehicles in each frame, the system estimates traffic density. This project demonstrates the application of basic AI concepts in solving real-world problems like traffic monitoring.

Introduction

Traffic congestion is a major problem in urban areas, leading to delays and increased fuel consumption. Intelligent systems can help monitor and manage traffic effectively. This project introduces a simple traffic density analyzer using computer vision techniques. It processes video input, detects vehicles, and classifies traffic density, providing a foundation for smart traffic management systems.

Problem Statement

Manual monitoring of traffic is inefficient and time-consuming. There is a need for an automated system that can:

- Detect vehicles in real-time
- Count vehicles accurately
- Classify traffic density

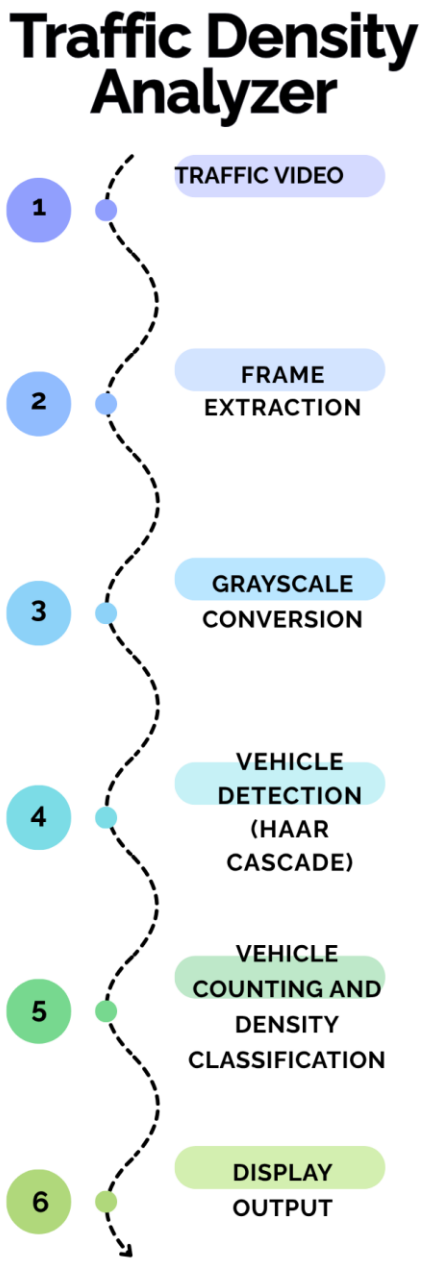
Objectives

- To detect vehicles using computer vision
- To count the number of vehicles in a video
- To classify traffic density as Low, Medium, or High
- To demonstrate AI concepts in a practical application

Methodology

The system follows these steps:

1. Load traffic video
2. Convert each frame to grayscale
3. Use Haar Cascade classifier to detect vehicles
4. Filter unwanted detections
5. Count valid vehicles
6. Classify traffic density based on count
7. Display results on screen



Tools and Technologies Used

- **Programming Language:** Python
- **Libraries:** OpenCV, NumPy
- **Development Tool:** Visual Studio Code
- **Version Control:** GitHub

System Design

Input:

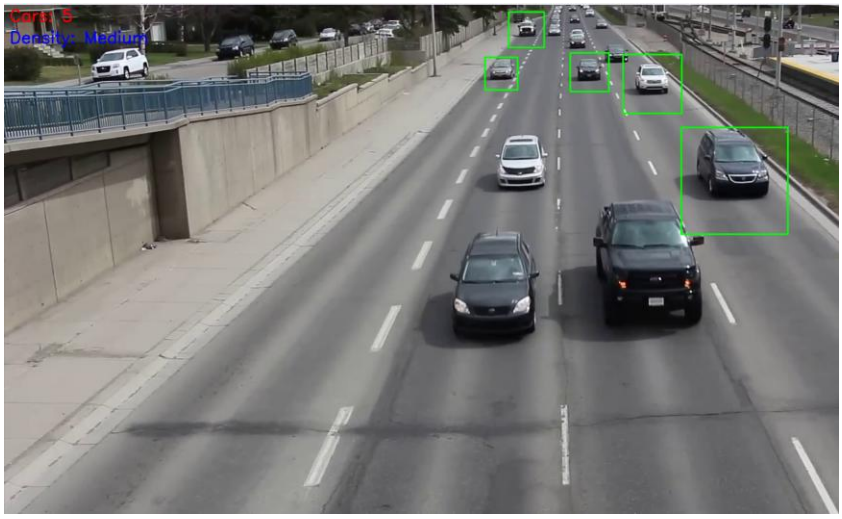
- Traffic video (traffic.mp4)

Process:

- Frame extraction
- Vehicle detection
- Counting
- Classification

Output:

- Video with bounding boxes
- Vehicle count
- Density level



Implementation

The project is implemented using Python and OpenCV. A pre-trained Haar Cascade classifier is used to detect vehicles in each frame. The system processes video frames in real-time and applies filtering techniques to improve detection accuracy. The number of detected vehicles is used to determine traffic density.

Results

The system successfully:

- Detects vehicles using bounding boxes
- Counts vehicles in each frame
- Classifies traffic density

Example output:

- Cars: 3 → Medium
- Cars: 1 → Low
- Cars: 8 → High

Advantages

- Simple and easy to implement
- Works with basic hardware
- Demonstrates real-world AI application
- Low computational cost

Limitations

- Haar Cascade may produce false detections
- Not very accurate for small or distant vehicles
- Performance depends on video quality

Future Scope

- Use deep learning models like YOLO
- Improve detection accuracy
- Real-time camera integration
- Smart traffic signal automation

Conclusion

This project demonstrates how AI and machine learning concepts can be applied to solve real-world problems such as traffic monitoring. Although the system uses basic techniques, it provides a foundation for more advanced intelligent traffic systems.

References

- OpenCV Documentation
- Python Documentation
- GitHub repositories for Haar Cascade
- www.youtube.com/@rileykearl (for traffic.mp4)